

Augmenting Cross-View Geo-Localization with Spatial Semantics from Vision Foundation Models

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Abstract

Cross-view geo-localization (CVGL) establishes correspondences between ground-level and satellite images of the same geographic location, serving as a fundamental technology for smart city applications, including autonomous navigation, urban planning, and location-based services. Current CVGL approaches fall into two categories: feature-based methods achieve superior performance through 2D representation learning but lack interpretability. Spatial-based methods provide geometric understanding and interpretable matching but suffer from limited spatial modeling and weak cross-view alignment, leading to lower performance. We reformulate CVGL from a spatial perspective and propose an auxiliary task-enhanced network. The network captures spatial semantics and provides explicit alignment processes with visualizable results. We introduce an auxiliary spatial semantic alignment (SSA) task that learns spatial structure via vision foundation models (VFM) and BEV transformation to enhance the primary CVGL task. The primary task captures visual semantics, including texture and appearance. Within this unified framework with shared encoders, the primary task enriches the learned embeddings by fusing spatial structure with visual semantics, yielding spatially complete representations. Extensive experiments on three standard CVGL benchmarks demonstrate that our method significantly surpasses previous spatial-based approaches while maintaining competitive performance with state-of-the-art (SOTA) feature-based methods, achieving 98.48% R@1 on CVUSA and 71.05% R@1 on CVACT_test. We provide comprehensive analyses through pixel-level activation maps and feature-space UMAP visualizations to validate both effectiveness and interpretability.

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CCS Concepts

• **Computing methodologies** → **Visual content-based indexing and retrieval**; *Matching*.

Keywords

Cross-view geo-localization, spatial semantic alignment, vision foundation models

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1 Introduction

Cross-view geo-localization (CVGL) is a computer vision task that aims to establish correspondences between ground-level images captured by street-view cameras and satellite/aerial images of the same geographic location. CVGL supports autonomous navigation, urban planning, and location-based services, including scenarios where GPS is unavailable or degraded. In dense urban environments, Global Navigation Satellite System (GNSS)-based positioning accuracy is significantly degraded due to signal blockage, severe multipath effects, and limited satellite visibility, resulting in unreliable or inaccurate localization[9]. CVGL is therefore required as a complementary solution. The primary challenge in CVGL lies in the significant viewpoint differences between ground-level and satellite images[6, 12, 25, 31]. Ground-level images capture street views with limited field of view and perspective distortion, while satellite images provide overhead perspectives with different scales and orientations. This fundamental difference makes it extremely difficult to establish reliable correspondences between these two modalities, as the same geographic location appears completely different from these contrasting viewpoints. To address this challenge, current methods can be categorized into two main types: feature-based and spatial-based approaches. Both follow the common encoder

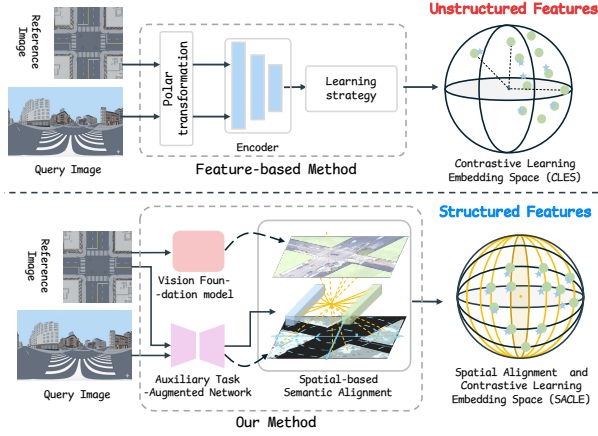


Figure 1: Feature-based methods encode satellite and ground images into high-dimensional vectors, yielding embeddings that lack interpretability and explicit spatial structure. Our framework uses a vision foundation model to extract satellite-view spatial semantic representations and aligns ground-view features to them, producing embeddings with explicit spatial alignment and higher interpretability.

+ similarity computation paradigm, which extracts features from ground and satellite images, compresses them into embeddings, and computes similarity between embeddings. However, they differ in their focus. Feature-based methods focus on learning 2D visual representations through three strategies. The first enhances the encoder by incorporating attention mechanisms and Transformer-based architectures [33, 34, 39, 41, 43]. The second employs polar transformations [5, 11, 30] to reduce perspective variations. The third improves the learning strategy by leveraging unlabeled data and hard negative sampling techniques [3, 10] to enhance performance. While these methods achieve strong performance, they lack spatial understanding capability and interpretability [29], which limits their application in scenarios requiring explainable decision-making and spatial reasoning.

In contrast, spatial-based methods focus on spatial semantic representation learning for cross-view matching. These methods recognize that the spatial layout and semantic configuration of scenes provide structural clues and reduce matching ambiguity caused by viewpoint differences. Unlike feature-based methods that primarily rely on appearance similarity, spatial-based methods explicitly model spatial relationships and geometric structures. For instance, SSA-Net [38] proposes a spatial-scale attention module to highlight correspondence layout features at different scales, where the encoded features represent both objects and their spatial locations. This enables the model to capture not only what objects are present but also where they are located relative to each other. Similarly, SAFA [24] employs spatial attention mechanisms to bring corresponding spatial features closer in the embedding space, facilitating better alignment between ground and satellite views. Additionally, GeoDTR [37] leverages spatial transformations and geometric understanding to establish cross-view correspondences by modeling the geometric relationships between views.

These approaches show that incorporating spatial semantic understanding into feature representation leads to more interpretable and geometrically consistent representations for CVGL.

However, these spatial semantic approaches face several fundamental design problems. First, the spatial relationship modeling is often limited to local geometric relationships rather than comprehensive semantic understanding across viewpoints. Second, these methods lack explicit mechanisms to align spatial semantic representations between ground and satellite views. Third, the spatial semantic information is not effectively integrated with the primary matching objective, leading to a disconnect between spatial understanding and localization accuracy.

An auxiliary task-enhanced network is proposed to tackle these limitations. As shown in Fig.1, feature-based methods rely solely on visual similarity learning, while our network integrates a spatial semantic alignment (SSA) auxiliary task with contrastive learning. The SSA task supervises the encoder to align ground-view spatial representations with satellite-view spatial representations, enabling cross-view spatial correspondence learning in pixel space. Ground-view representations are obtained through a bird’s-eye view (BEV) transformation task, while satellite-view representations are generated by vision foundation models (VFM). This auxiliary task provides geometric understanding to guide the encoder, while contrastive learning ensures discriminative embeddings for matching. Through this dual-task paradigm, the network produces embeddings with both spatial structure and visual discriminability.

The main contributions of this paper include:

- (1) We propose an auxiliary task-enhanced network for CVGL, where an auxiliary SSA task enhances the primary cross-view retrieval objective. SSA learns spatial structure by aligning satellite spatial semantic representation with ground spatial semantic representation in pixel space, providing explicit geometric guidance.
- (2) We design the SSA objective to align ground-view spatial semantic representation with satellite-view spatial semantic representation in pixel space. The ground-view spatial semantic representation is obtained via BEV transformation, while the satellite-view spatial semantic representation is generated by VFM. The alignment enables shared encoders to learn consistent scale, orientation, and morphology across views.
- (3) Extensive experiments on three standard CVGL benchmarks show state-of-the-art (SOTA) performance among spatial-based methods, 98.48% R@1 on CVUSA and 71.05% R@1 on CVACT_test, while remaining competitive with the best feature-based method. We further provide pixel-level activation maps and feature visualizations to validate effectiveness and analyze limitations.

2 Related Work

BEV transformation for spatial semantic alignment. BEV transformation is a pivotal technology in autonomous driving. It has traditionally relied on LiDAR point clouds for precise spatial information. However, LiDAR-based approaches are often costly and provide relatively sparse semantic information. Consequently, recent research has shifted towards exploring vision-based BEV perception methods. While images offer rich semantic content, they present challenges due to the inherent lack of depth information. To address this, many efforts have focused on estimating depth from

images and transforming camera-view features into the BEV space [15, 19, 21, 28]. Despite significant progress, these image-based BEV methods generally still lag behind their LiDAR-based counterparts in terms of spatial perception accuracy and robustness. Given our research objective of leveraging BEV perception to improve geospatial information extraction from image pairs in cross-view tasks, our work predominantly draws inspiration from techniques for direct BEV representation generation from images, complemented by an examination of methods employing generative models to synthesize aerial/satellite or BEV images from ground imagery [1, 35].

Feature-based vs. spatial semantics in CVGL. Mainstream CVGL learns view-invariant 2D embeddings with an encoder and contrastive retrieval. Stronger backbones and attention/Transformer modules improve discrimination and robustness [33, 34, 39, 41, 43], and architectural refinements such as LPN [27] and SAIG-D [43] yield further gains. Hard-negative mining and curriculum-style sampling, e.g., Sample4Geo [3], substantially boost retrieval but still operate within purely feature-based paradigms. To mitigate viewpoint gaps, polar or log-polar transforms [5, 11, 30] partially normalize geometry but may distort boundaries and lose detail. In contrast, spatial-focused methods encode layout and relative configuration to curb ambiguity: SAFA applies spatial attention on polar grids [24], SSA-Net introduces spatial-scale attention for layout-aware features [38], and GeoDTR leverages geometric transformations for cross-view consistency [37]. Despite progress, most spatial approaches lack a unified and complete spatial semantic representation and couple spatial reasoning only implicitly to the retrieval loss, leaving alignment under-constrained and interpretability limited.

3 Method

As shown in Fig. 2, we propose an auxiliary task-enhanced framework integrating two complementary objectives through shared encoders. The auxiliary spatial semantic alignment objective aligns ground-view and satellite-view spatial semantic representations in pixel space, providing geometric supervision. The primary retrieval objective learns discriminative descriptors from encoder features enriched by this alignment. Joint optimization fuses spatial structure with visual semantics, yielding geometrically grounded, interpretable, and discriminative representations for CVGL.

3.1 Unified Spatial-Semantic Representation and Alignment for CVGL with VFM

3.1.1 Spatial Semantics Construction. We encode spatial semantics as a graph G where landmarks are nodes (e.g., buildings, observation targets) and edges capture relative geometry (distances, orientations). In ground view, detected landmarks form $P_q = \{\mathbf{p}_0, \dots, \mathbf{p}_M\}$, with observation $\mathbf{p}_0 = (x_0, y_0)^\top$ and references $P = [\mathbf{p}_1, \dots, \mathbf{p}_M]^\top$. Each landmark carries appearance attributes $\mathbf{a}_i \in \mathbb{R}^d$ (shape/texture/size), yielding node features $X_q = [\mathbf{a}_1, \dots, \mathbf{a}_M]^\top$. We infer geometry from RGB with noise increasing by depth. The ground representation is $G_q = (P_q, W_q, E_q)$, while satellite view $G_r = (P_r, W_r, E_r)$ excludes the observation point and offers more reliable weights.

However, graph-based matching faces critical challenges. Single RGB images provide incomplete observations due to occlusion, scale change, and depth ambiguity. Cross-view comparability is problematic: the ground image contains an observer node with a limited

field of view and view-dependent topology, while the satellite-view is global and normalized. Direct graph matching becomes combinatorial and ill-posed without reliable anchors. To resolve this, we project each view’s spatial graph into a pixel coordinate system as binary masks preserving landmark shape, orientation, and distance. This pixel-space representation enables differentiable alignment via supervised learning, avoiding combinatorial graph matching while maintaining spatial structure.

3.1.2 Satellite-View Spatial Semantic Representation. To obtain satellite-view spatial semantic representation G_q^r in pixel coordinates, we leverage a VFM rather than learning from scratch. Learning spatial semantic representations purely through task-specific networks is unreasonable given the limited supervision signals in CVGL datasets and the complexity of spatial semantic understanding. VFMs pre-trained on massive diverse data offer powerful semantic segmentation capabilities that provide reliable spatial priors. We employ SAMGeo [32], a VFM tailored for remote sensing, to generate per-class binary masks. We focus on buildings and tall trees as they provide stable spatial cues across views. Since SAMGeo’s segmentation is sensitive to prompt synonyms (Fig. 3), we apply the **Multi-source Fusion Strategy** to combine predictions from multiple prompts, enhancing mask quality. Details are in A.1.

3.1.3 Ground-View Spatial Semantic Representation. To better align with the satellite-view spatial semantic representation G_q^r , we generate the ground-view spatial semantic representation P_q in pixel coordinates by transforming ground image features into BEV following the Lift-Splat-Shoot paradigm [1, 18]. Given features $F_q \in \mathbb{R}^{C \times H \times W}$ extracted by a backbone encoder, we first apply dynamic polar projection. The dynamic weights $\alpha = \text{Softmax}_D(W_{\text{ctx}}(F_q))$ are predicted by a context network W_{ctx} , where D denotes the discretized radial (depth) dimension. We lift features into a frustum via $F_{\text{lift}} = \alpha \odot F_q'$, then compress along height to obtain pseudo-polar features $F_{\text{polar}} = \sum_{h=1}^H F_{\text{lift}}(:, \cdot, h, \cdot) \in \mathbb{R}^{C \times D \times W}$, where W corresponds to the angular dimension. Next, F_{polar} is resampled into a BEV Cartesian grid $G \in \mathbb{R}^{S \times S \times 2}$ using a differentiable bilinear sampler [7], yielding $F_{\text{BEV}} = S(F_{\text{polar}}, G) \in \mathbb{R}^{C \times S \times S}$. Here, S denotes the spatial resolution of the BEV grid, and we set $S = 48$ in all experiments. The operator S represents bilinear interpolation-based sampling, which enables end-to-end differentiable transformation from polar to Cartesian coordinates. Finally, a transformation decoder $\mathcal{D}$ maps F_{BEV} to per-class logits $P_q \in \mathbb{R}^{K \times S \times S}$, which are used for spatial semantic alignment (SSA).

3.1.4 Spatial Semantic Alignment (SSA). Satellite and ground spatial semantic representations share the same pixel coordinate frame but are not aligned. Scale, global orientation, and landmark size and shape may differ, and the ground-view spatial semantic representation lacks direct supervision. We use the satellite-view spatial semantic representation as the training signal and supervise the ground-view representation to match it. This removes the need for explicit graph matching and lets the network learn consistent scale, orientation, and morphology in pixel coordinates. We optimize the alignment with cross-entropy $\mathcal{L}_{\text{ce}}$ and Dice $\mathcal{L}_{\text{dice}}$, as defined in Eq. 1, yielding consistent pixel space semantics that support CVGL:

$$\mathcal{L}_{\text{align}}(P_q, G_q^r) = \mathcal{L}_{\text{ce}}(P_q, G_q^r) + \mathcal{L}_{\text{dice}}(P_q, G_q^r) \quad (1)$$

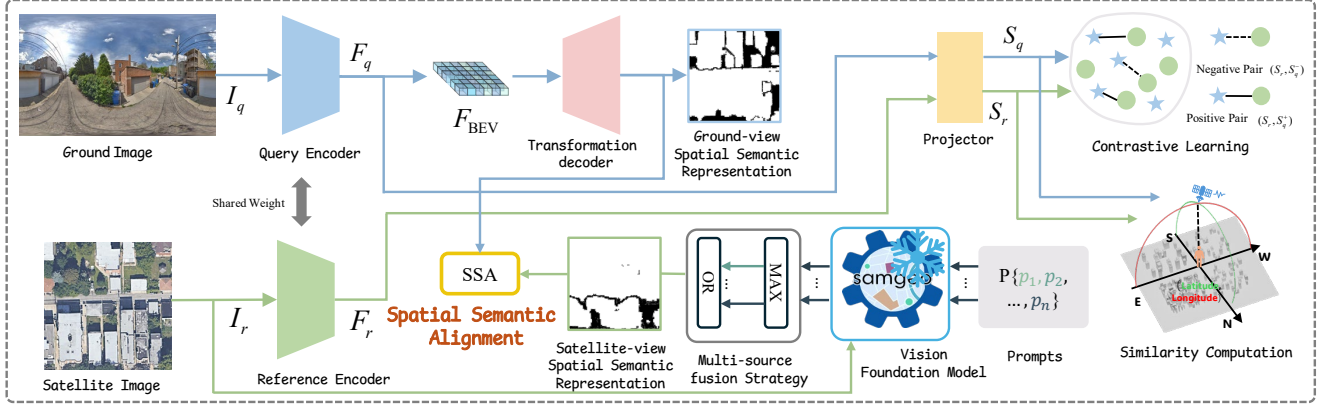


Figure 2: Overview of the framework. Shared encoders $\mathcal{E}_q$ and $\mathcal{E}_r$ extract features from ground and satellite images, and a projector $\mathcal{P}$ maps them to global descriptors for retrieval. Transformation decoder $\mathcal{D}$ transforms ground features with to produce the ground-view spatial semantic representation P_q and derives the satellite-view representation G_q^r from SAMGeo using the Multi-source Fusion Strategy. Pixel-space alignment supervises P_q to match G_q^r , and joint optimization with contrastive learning yields spatially grounded, discriminative embeddings.

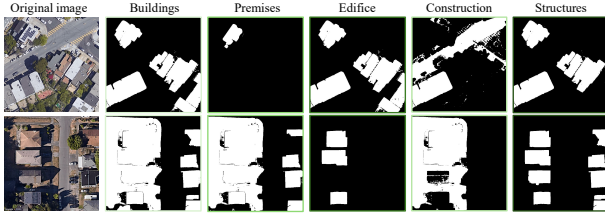


Figure 3: On VIGOR, five building synonyms produce distinct SAMGeo masks, demonstrating prompt sensitivity and motivating the Multi-source Fusion Strategy.

3.2 Auxiliary Task-Enhanced Framework

Our framework integrates two objectives through shared encoders: a primary objective learns visual features for CVGL, and an auxiliary SSA objective provides spatial semantic supervision. This design addresses the challenge of establishing spatial correspondences across different viewpoints while maintaining visual discriminability.

The SSA objective operates in pixel coordinate space, aligning ground-view spatial semantic representation obtained with satellite-view spatial semantic representation. The shared encoders learn consistent scale, orientation, and morphology across views. The primary retrieval objective uses encoder features enriched by SSA to compute similarity between image descriptors. Joint optimization fuses spatial structure with visual semantics, producing geometrically grounded and discriminative representations.

Given ground and satellite images, shared encoders extract features that a projection maps to compact global descriptors. We compute similarity with cosine similarity or inner product and optimize the descriptors with the InfoNCE objective:

$$\mathcal{L}_{cl} = -\log \frac{\exp(S_q \cdot S_r^+ / \tau)}{\sum_k \exp(S_q \cdot S_{r_k} / \tau)} \quad (2)$$

where $S_q \in \mathbb{R}^d$ denotes the descriptor of a ground image, $S_r^+ \in \mathbb{R}^d$ is the positive satellite descriptor of the matched satellite image, and $\{S_{r_k}\}_{k=1}^N$ are negative satellite descriptors sampled from other locations. We compute similarity using cosine similarity. We jointly optimize the contrastive objective with SSA, and the overall training loss is defined as:

$$\mathcal{L}(S_r, S_q, P_q, G_q^r) = \mathcal{L}_{cl}(S_r, S_q) + \frac{w}{2} \mathcal{L}_{align}(P_q, G_q^r) \quad (3)$$

where w is the task weight for the SSA loss.

4 Experiment

4.1 Experimental Setups

Datasets. We evaluate on three standard CVGL benchmarks and one auxiliary dataset with distinct roles: CVUSA and CVACT provide geometrically rectified, aligned settings to assess performance under favorable geometry; VIGOR, without rectification and with same-/cross-area splits, stresses robustness to geometric asymmetry and domain shift; VIGORv2 augments VIGOR with center-aligned aeriels and BEV layout maps, serving analysis of our SSA behavior and as a source of satellite-view spatial semantic representation.

CVUSA[30] consists of 44,416 image pairs collected across the United States, with 35,532 pairs for training and 8,884 for testing. Ground-level images (224×1,232) and satellite images (750×750) are geometrically rectified with geographic north aligned at the top center, ensuring consistent spatial orientation across the dataset.

CVACT[14] contains 137,218 image pairs collected in Canberra, Australia, divided into 35,532 training pairs, 8,884 validation pairs, and 92,802 testing pairs. It offers higher resolutions with ground-level images at 832×1664 pixels and satellite images at 1200×1200 pixels. Like CVUSA, CVACT employs geometric rectification using sensor metadata to align imagery with geographic north.

VIGOR[41] comprises 90,618 satellite images and 105,214 ground-level panoramic images from four U.S. cities (New York, Seattle, San Francisco and Chicago). Satellite images are 640×640 pixels, and

ground images are 1024×2048 pixels. Unlike CVUSA and CVACT, VIGOR lacks geometric rectification. The dataset supports two evaluation protocols: Same-area (all cities for training/testing) and Cross-area (two cities for training, two for testing). Another novelty is the introduction of semi-positive images. For each image pair, three satellite-view neighbors also cover regions of the street-view image, yet their capture locations lie outside the central region.

VIGORv2[1] extends VIGOR with center-aligned aerial images and additional modalities including BEV layout maps and textual descriptions. The BEV maps are generated using OpenStreetMap API with eight semantic categories (buildings, roads, water bodies, etc.) rendered at 300×300 resolution. We utilize these BEV layouts as spatial priors for analysis and experimentation.

Evaluation metric. In evaluating the VIGOR, CVUSA, and CVACT datasets, we employed the widely adopted Top-K Recall (Recall@K) metric to assess model performance, specifically calculating Recall@K for K values of 1 and 5, as well as evaluating performance at the 1% candidate image threshold. Additionally, for the VIGOR, Hit Rate is utilized as a supplementary metric to provide a more comprehensive evaluation of model efficacy.

In the visualization results, the symbols (“✓”, “✗”) indicate whether a method produces visual outputs that are intrinsically derived from its model design and provide human-interpretable explanations of spatial semantics and alignment. A “✗” does not imply the absence of attention mechanisms, but rather that the visualizations are not directly grounded in the model’s learned representations.

Hyperparameters. Experiments run on 4 NVIDIA 4090D GPUs. The shared query and reference encoders use the first four stages of ConvNeXt-Base[16]. The projector uses the remaining stages to produce global descriptors. We remove the terminal normalization and classification head and prune one intermediate block in the second stage to control depth and computation. This yields a stable shared encoder for both views. The projector reuses a late ConvNeXt block with the original normalization and pooling path. Satellite images are downsampled to 384×384 and ground images to 384×768. We train for 100 epochs with AdamW[17] and a cosine schedule. The learning rate decays from 1×10^{-3} to 1×10^{-4} . The SSA task weight w is 1 and the batch size is 32. For contrastive sampling, we adapt the original Dynamic Similarity Sampling[3] to our setting. This adaptation is motivated by our change in the image representation learning paradigm, and details appear in A.2. Our implementation is publicly available¹

4.2 Results and Discussions

4.2.1 Performance Comparison with Baselines. Our model achieves SOTA performance among spatial-based methods across all datasets. On CVUSA and CVACT_test, we achieve 98.48% R@1 and 71.05% R@1 respectively, significantly outperforming existing spatial-based methods like GeoDTR by 3.05% and 6.48% percentage points, and maintaining competitive performance with only a 0.20% gap compared to the best feature-based method Sample4Geo on CVUSA. On the VIGOR dataset, our method achieves 55.16% R@1 on the SAME split, competitive with other spatial-based approaches but showing a significant performance gap compared to Sample4Geo, with disparities of 22.70% on SAME split and 39.09% on CROSS split.

This performance variation across datasets highlights the critical impact of geometric alignment on SSA effectiveness.

The visualization results provide crucial insights into these performance patterns. As illustrated in Fig.4, the visualization analysis reveals stark differences in spatial semantic learning quality across datasets. On CVUSA and CVACT datasets, our model learns effective spatial semantic representations that successfully capture landmark orientations, distances, and contours, with ground-view spatial semantic representation showing recognizable building structures and coherent spatial layouts that correspond well to satellite imagery. In contrast, on the VIGOR dataset, the learned spatial semantic representations are severely degraded, exhibiting chaotic patterns with no discernible spatial structure, which directly explains the poor quantitative performance observed. This visualization evidence demonstrates the critical importance of center alignment and directional alignment for our method’s spatial semantic alignment task. The center alignment ensures consistent spatial reference points between ground and satellite views, while directional alignment provides reliable orientation cues that enable meaningful spatial correspondence learning. Without these geometric alignment conditions, the SSA becomes ill-defined and produces erroneous spatial representations that interfere with rather than enhance the matching process. To validate the importance of spatial semantic representation quality, we conduct an experiment using VIGORv2’s layout maps as satellite spatial semantic representations for SSA task training, where building and tree labels serve as landmarks. As shown in the Table.3, Ours+VIGORv2 achieves notable improvements with 58.55% R@1 on SAME split and 26.19% R@1 on CROSS split. These results demonstrate that the precision and definition of satellite spatial semantic representations significantly impact final performance, with detailed analysis provided in the discussion and A.4.

4.2.2 Ablation Study. Table.2 shows that SSA effect varies across datasets with different geometric alignment. On CVUSA and CVACT with geometric rectification, our model with SSA matches pure feature-based methods (w/o SSA): R@1 reaches 98.48% vs 98.50% on CVUSA and 90.05% vs 89.91% on CVACT_val. This demonstrates our spatial semantic representation design is effective. Geometric alignment provides anchor points for BEV transformation learning, enabling SSA to learn spatial representations that match feature-based performance. On VIGOR (SAME) without geometric rectification, SSA causes performance degradation: R@1 drops from 68.54% to 59.72%—an 8.82% decrease. Spatial semantic alignment depends on geometric correspondence between image pairs. On VIGOR, lack of geometric rectification makes spatial alignment ill-defined and causes feature space collapse. This introduces optimization conflicts instead of useful guidance. Detailed analysis is provided by the two experiments that follow: pixel-level activation maps and feature space analysis visualization via uniform manifold approximation and projection (UMAP).

4.2.3 Interpretability Analysis via Pixel-Level Activation Maps. As shown by the activation maps in Fig.5, adding the spatial semantic alignment objective yields minimal change in R@K metrics but produces substantial differences in pixel-level attention, demonstrating that our model successfully learns meaningful spatial semantic representations. Without spatial semantic guidance, the

¹https://github.com/q5756578/SSA_for_CVGL

Table 1: Performance comparison on the CVUSA and CVACT. * indicates the model used polar transformation methods

Type	Approach	CVUSA				CVACT_Val				CVACT_Test				Vis. Result
		R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%	
Feature-based	LPN [27]	85.79	95.38	96.98	99.41	79.99	90.63	92.56	–	–	–	–	–	✗
	LPN*	92.83	98.00	98.85	99.78	79.99	90.63	92.56	–	–	–	–	–	✗
	TransGeo [39]	94.08	98.36	99.04	99.77	84.95	94.14	95.78	98.37	–	–	–	–	✗
	DSM [24]	91.96	97.50	98.54	99.67	82.49	92.44	93.99	97.32	35.63	60.07	69.10	84.75	✗
	SAIG-D [43]	96.08	98.72	99.22	99.86	89.21	96.07	97.04	98.74	–	–	–	–	✗
	SAIG-D*	96.34	99.10	99.50	99.86	89.06	96.11	97.08	98.89	67.49	89.39	92.30	96.80	✗
	L2LTR [34]	91.99	97.68	98.65	99.75	83.14	93.84	95.51	98.40	58.33	84.23	88.60	95.83	✗
	L2LTR*	94.05	98.27	98.99	99.67	84.89	94.59	95.96	98.37	60.72	85.85	89.88	96.12	✗
	Sample4geo	98.68	99.68	99.78	99.87	90.81	96.74	97.48	98.77	71.51	92.42	94.45	98.70	✗
Spatial-based	GeoNet-II [42]	58.9	81.8	88.3	98.7	–	–	–	95.8	–	–	–	–	✗
	SSA-Net [38]	91.96	97.69	98.57	99.71	84.23	94.33	95.95	98.36	–	–	–	–	✗
	SAFA* [24]	89.84	96.93	98.14	99.64	81.03	92.80	94.84	–	–	–	–	–	✗
	GeoDTR [37]	93.76	98.47	99.22	99.85	85.43	94.81	96.11	98.26	62.96	87.35	90.70	98.61	✗
	GeoDTR*	95.43	98.86	99.34	99.86	86.21	95.44	96.72	98.77	64.52	88.59	91.96	98.74	✗
	Ours	98.48	99.56	99.61	99.85	90.05	96.08	96.78	98.35	71.05	91.55	93.68	98.31	✓



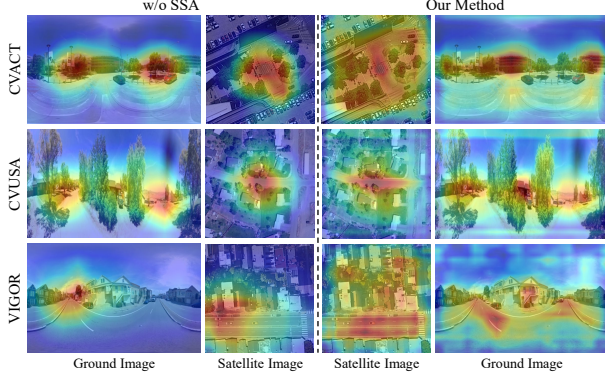
Figure 4: Visualization of results from three datasets. We use azimuth degrees to represent the orientation changes of images captured from different viewpoints, illustrating directional information. In our image setup, the north direction is defined as 0°, with the left side mapped to -180° and the right side to 180°. This centered azimuth definition effectively captures image orientation and is particularly suitable for tasks involving directional alignment or view normalization.

model focuses on scattered high-texture regions, making it difficult to interpret what features truly drive CVGL decisions. In contrast,

with alignment, the satellite branch attends to globally coherent spatial layouts, and the ground branch emphasizes landmark regions consistent with our design, confirming that the learned features

Table 2: Ablation study on SSA. w/o SSA denotes the variant without the spatial semantic alignment task.

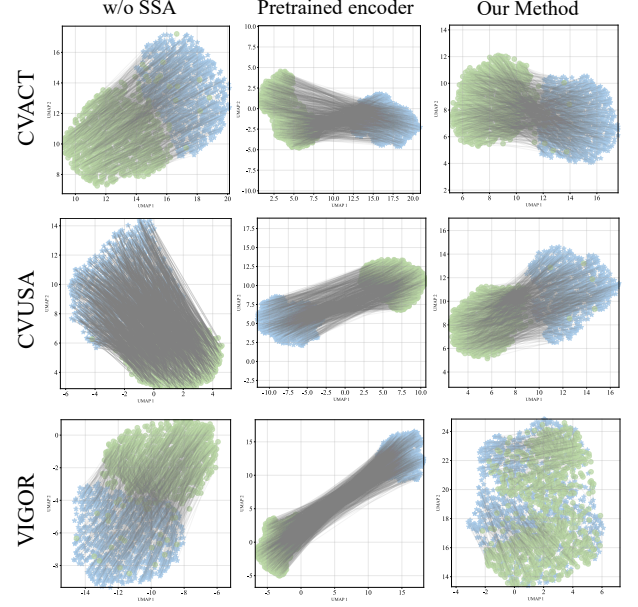
Method	CVUSA			CVACT_val			VIGOR(SAME)			
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	Hit Rate
w/o SSA	98.50	99.68	99.81	89.91	96.23	97.45	68.54	89.26	96.10	78.46
Ours	98.48	99.56	99.61	90.05	96.08	96.78	59.72	88.71	93.46	72.12

**Figure 5: Qualitative comparison of pixel-level activation maps. Activation patterns differ markedly in where the model attends on satellite and ground views.**

encode scene configuration rather than incidental textures. The effectiveness of spatial semantic learning is fundamentally dependent on geometric alignment between image pairs, as evidenced by the stark performance differences between geometrically aligned datasets and unaligned ones. On aligned datasets, our model establishes reliable cross-view spatial correspondences and learns consistent spatial semantics, while on VIGOR, the lack of geometric alignment makes spatial alignment ill-defined, producing unusable BEV targets and optimization conflicts.

Our analysis further reveals that spatial semantic representations from ground and satellite views are inherently non-equivalent, necessitating careful consideration of their different information capacities. Ground images provide richer local spatial information due to their proximity to landmarks and detailed perspective of spatial relationships, but are inherently limited by their restricted field of view, capturing only partial spatial semantic representations of the scene. In contrast, satellite images offer comprehensive global spatial coverage but lack fine-grained local details such as object textures. This fundamental asymmetry between modalities explains why geometric alignment is crucial for effective spatial semantic alignment: Geometric alignment bridges this asymmetry by providing anchor points that establish spatial correspondences between views, enabling the BEV transformation to learn consistent cross-view mappings. Without geometric alignment, the model cannot determine region correspondences and prevent meaningful spatial semantic learning.

4.2.4 Feature Space Analysis via UMAP. To better understand how spatial semantic alignment affects the learned feature representations, we conduct a high-dimensional feature space analysis using

**Figure 6: UMAP visualization of feature distributions. We randomly sample 1000 image pairs and project their high-dimensional embeddings to 2D space using UMAP. ● denotes satellite images and ★ denotes ground-level images.**

UMAP dimensionality reduction. UMAP is a manifold learning technique that projects high-dimensional embeddings into 2D space while preserving the global structure of the data. As shown in Fig. 6, the feature distributions reveal counterintuitive patterns across different datasets. Starting from pretrained encoder weights, the two modalities exhibit substantial separation in the feature space, with inter-modality distances far exceeding intra-pair distances. Our model appears to prioritize reducing inter-pair distances over maintaining intra-class separation, leading to results where better feature mixing paradoxically correlates with worse performance.

Most strikingly, on the VIGOR dataset where our model performs worst, satellite and ground-level images become almost completely mixed in the feature space—a distribution that would theoretically be ideal for CVGL but coincides with the poorest retrieval performance. We attribute this phenomenon to feature space collapse, where the lack of geometric alignment in VIGOR creates insurmountable difficulties for spatial semantic alignment, making the alignment task ill-defined and significantly increasing learning difficulty. Since our model employs shared-weight encoders for both modalities, the optimization conflicts arising from misaligned spatial supervision directly propagate through the shared parameters, causing the feature space to collapse. This results in the overlapping distributions observed in the UMAP visualization, where the model sacrifices discriminative power in favor of reducing matching distances, ultimately degrading retrieval performance despite achieving seemingly ideal feature mixing. In contrast, purely feature-based methods effectively avoid this issue.

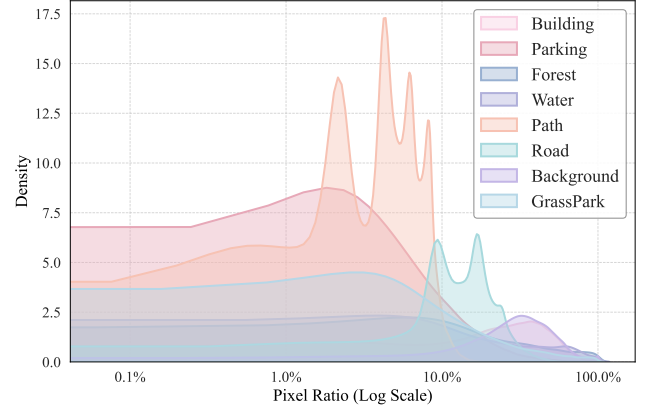
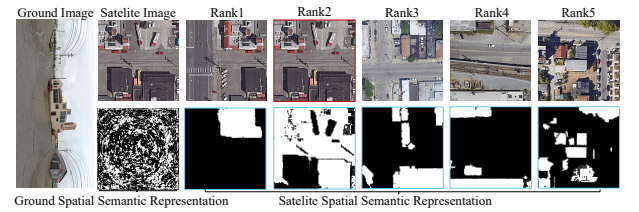
Table 3: Performance comparison between our approach and representative baselines on VIGOR

Type	Approach	SAME					CROSS					Vis. Result
		R@1	R@5	R@10	R@1%	Hit Rate	R@1	R@5	R@10	R@1%	Hit Rate	
Feature-based	Siamese-VGG[40]	18.69	43.69	55.36	97.55	21.96	2.77	8.61	12.94	62.64	3.16	✗
	SAFA+Mining[23]	38.02	62.87	71.12	97.63	41.81	9.23	21.12	28.02	77.84	9.92	✗
	VIGOR[41]	41.07	65.81	74.05	98.37	44.71	9.23	21.12	28.02	77.84	9.92	✗
	TransGeo[39]	61.48	87.54	91.88	99.56	73.09	18.99	38.24	46.91	88.94	21.21	✗
	Sample4geo[3]	77.86	95.66	97.21	99.61	89.92	61.70	83.50	88.00	98.17	69.87	✗
Spatial-based	SAFA[41]	33.93	58.42	68.12	98.24	36.87	8.20	19.59	26.36	77.61	8.85	✗
	SAIG-D[43]	65.23	80.88	–	99.68	74.11	33.05	55.94	–	94.64	36.71	✗
	GeoDTR [37]	56.51	80.37	86.21	99.25	61.76	30.02	52.67	61.45	94.40	30.19	✗
	GeoDTR+[36]	59.01	81.77	87.10	99.07	67.41	36.01	59.06	67.22	94.95	39.40	✗
	Ours	55.16	84.72	90.69	99.75	66.13	22.61	45.92	55.76	94.18	26.49	✓
	Ours+VIGORv2	59.72	88.71	93.46	99.84	72.72	31.50	58.33	67.74	96.95	38.01	✓

Table 4: Performance comparison on CVACT with CVUSA-trained models. * indicates models use polar transformation.

Method	CVUSA → CVACT-Val				CVUSA → CVACT-Test			
	R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%
L2LTR [34]	47.55	70.58	–	91.39	–	–	–	–
L2LTR*	52.58	75.81	77.39	93.51	–	–	–	–
GeoDTR [37]	47.79	70.52	–	92.20	11.24	18.69	23.67	72.09
GeoDTR*	53.16	75.62	81.90	93.80	22.09	32.22	39.59	85.53
Ours	46.40	70.34	77.45	93.25	21.45	43.43	52.13	93.34

4.2.5 Generalisation Performance. To further investigate the generalization capability of our model, we conduct cross-dataset evaluation on the CVUSA and CVACT datasets, where the model is trained on the training split of CVUSA and subsequently evaluated on the validation and test splits of CVACT. Our experimental results reveal that the model exhibits limited generalization performance in cross-dataset settings, with a notable gap compared to single-task baselines. This phenomenon can be attributed to the inherent limitations of our spatial semantic alignment task, which utilizes shared encoders with limited representational capacity for learning domain-invariant spatial semantic features. As shown in Table.4, when trained on CVUSA and evaluated on CVACT, our method achieves 46.40% R@1 on CVACTVal and 21.45% R@1 on CVACT-Test, demonstrating weaker cross-dataset generalization compared to feature-based approaches that rely on more robust global descriptors. The SSA task’s dependency on consistent geometric correspondences between ground and satellite views makes it particularly vulnerable to domain shifts in architectural styles, building layouts, and geographic contexts between different datasets. However, our method shows competitive performance on higher recall metrics, achieving 43.43% R@5 and 52.13% R@10 on CVACTTest, outperforming existing spatial-based methods and demonstrating the substantial potential and importance of learning spatial semantic representations for cross-view matching.

**Figure 7: Class distribution of layout maps in the VIGORv2 dataset. We further analyze the variation in ground object category distributions across different cities.****Figure 8: Visualization of retrieval results on the VIGOR dataset. The figure shows the top-ranked satellite images retrieved for a given ground query, along with the corresponding spatial semantic representations.**

4.2.6 Discussion. To examine how different spatial-semantic priors influence the auxiliary task and, we conduct controlled experiments on the VIGOR dataset using VIGORv2 layout annotations with different foreground definitions. As shown in Table. 5, introducing

building+tree priors boosts R@1 from 50.84% to 59.72% on SAME and from 16.83% to 31.13% on CROSS, while using parking priors yields comparable gains of 60.34% and 30.87% R@1 respectively. Despite the markedly different foreground distributions in Fig. 7, the outcomes are similar, indicating that the fidelity and consistency of spatial supervision matter more than the specific class label used as foreground. Since our auxiliary objective is formulated as mask generation, both the number of categories and their pixel proportions directly shape optimization dynamics through class imbalance and foreground sparsity, thereby affecting the attainable performance. On current datasets these distributional constraints are inherent and difficult to avoid.

Additionally, as illustrated in Fig. 8, the VIGOR dataset contains near-neighbor sampling and lacks explicit geometric alignment, which further increases the optimization difficulty. Under these conditions, failures in spatial semantic learning can have adverse effects: when encountering visually similar ground structures, the incorrectly learned ground-side spatial semantic representations may dominate and degrade retrieval performance. We provide further analysis of these failure cases in A.4. Moreover, VIGOR’s perpendicular satellite imagery introduces irrelevant visual elements like building facades and directional shadows that disrupt spatial semantic alignment, further complicating the optimization process. Designing more effective spatial semantic representations remains a direction for future exploration. A promising approach is to move beyond flat masks toward hierarchical, structurally complete spatial semantic representations of places, capturing multi-scale regions such as block–parcel–building hierarchies, relational topology like road–parcel adjacency, and orientation cues to provide richer supervision for the primary CVGL objective.

Table 5: Retrieval performance under SAME and CROSS data splits with different spatial priors.

Data Split	Setting	R@1	R@5	R@10	R@1%	Hit Rate
SAME	Ours	50.84	81.95	88.81	99.77	61.03
	Building+tree	59.72	88.71	93.46	99.84	72.52
	All	58.55	88.06	92.84	99.84	71.38
CROSS	Ours	16.83	36.56	46.25	91.56	19.81
	Building+tree	31.13	58.45	67.56	97.03	37.48
	All	26.14	51.56	61.31	95.74	31.40

5 Limitations

While the proposed framework demonstrates strong performance and interpretability, it currently does not explicitly incorporate feature-based components. The framework is inherently extensible, and integrating such components may further improve both performance and interpretability. We leave this direction for future work.

6 Conclusion

This work explores the role of spatial semantics in advancing the design of CVGL methods. By incorporating spatial semantic alignment, we significantly enhance cross-view feature learning and scene understanding, achieving a performance breakthrough over existing

spatial-based approaches while improving model interpretability. However, our model still operates within the embedding-based retrieval paradigm, which faces inherent limitations in capturing complex spatial structures and hierarchical relationships[8, 26]. It also requires geometric alignment between image pairs. We believe spatial semantics holds greater potential for CVGL through better utilization of hierarchical structured data and graph-based representations[22] beyond flat embeddings. We also recognize the emerging potential of world models[2, 4] in learning spatial dynamics and vision-language models[13, 20] in understanding geometric and semantic relationships. How to effectively leverage these foundation models to build more structured, interpretable, and scalable CVGL systems remains an open question for developing next-generation methods. Beyond methodological contributions, our framework is designed to be deployment-friendly. It can be straightforwardly deployed on cloud platforms using existing infrastructure, and can be adapted to edge devices via model quantization and conversion techniques, enabling practical deployment in resource-constrained environments such as mobile platforms and autonomous systems.

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A Appendix

A.1 Multi-source Fusion Strategy.

Given a set of prompt words $P = \{p_1, p_2, \dots, p_n\}$ and satellite image I_r , the SAMGeo model $\mathcal{M}$ generates corresponding segmentation masks $G = \{G_{p_1}^r, G_{p_2}^r, \dots, G_{p_n}^r\}$, where each mask $G_{p_i}^r$ corresponds to prompt p_i . To obtain robust spatial semantic representations for satellite images, we leverage multiple prompts targeting the same semantic categories but with different linguistic expressions. We define semantic categories $C = \{c_1, c_2, \dots, c_j\}$, where each category represents a distinct object type relevant to spatial semantic alignment. The prompt set P is partitioned into category-specific subsets $P_j = \{p_i \in P | f(p_i) = c_j\}$, where f denotes the category mapping function. Correspondingly, the generated segmentation masks are grouped as $G_j = \{G_{p_i}^r \in G | f(p_i) = c_j\}$, enabling category-wise processing for robust spatial semantic representation generation.

To synthesize multiple segmentation masks into a unified spatial semantic representation for satellite images, we employ a two-stage fusion process. First, within each semantic category, we apply majority voting to select the most reliable mask from multiple prompt-generated candidates. Then, we combine all category masks through pixel-wise OR operations to produce the G_p^r :

$$G_{c_j}^r = \arg \max_{G_{p_i}^r \in G_j} \text{count}(G_{p_i}^r) \quad G_p^r = G_{c_1}^r \vee G_{c_2}^r \vee \dots \vee G_{c_j}^r \quad (4)$$

where $\arg \max$ count performs majority voting across pixel positions within each category to enhance reliability, and $\vee$ denotes pixel-wise OR operation for cross-category integration. This multi-source fusion strategy effectively aggregates complementary information from different prompts, yielding robust spatial semantic representations that serve as supervision signals for the SSA task.

In our experimental setup, we focus on two primary semantic categories that exhibit significant height in remote sensing imagery: buildings and trees, defining $C = \{\text{building}, \text{tree}\}$. Given the computational cost of SAMGeo (approximately two hours per prompt on VIGOR dataset), we carefully select a balanced prompt set {"artificial buildings", "structures", "trees"} to optimize the trade-off between segmentation quality and efficiency. This configuration partitions the masks into $G_{\text{building}} = \{G_{p_1}^r, G_{p_2}^r\}$ for building-related prompts and $G_{\text{trees}} = \{G_{p_3}^r\}$ for tree-related prompts. Through the aforementioned two-stage fusion process, we synthesize these multiple segmentation masks into a comprehensive spatial semantic representation G_p^r for each satellite image, which serves as the ground truth supervision for training the SSA task.

Fig.9 demonstrates the effectiveness of our multi-source fusion strategy on the VIGOR dataset. The synthesized spatial semantic representations successfully capture prominent buildings and trees from satellite imagery, providing rich spatial semantic information for SSA training. While our generated spatial semantic representations exhibit some inconsistency compared to the BEV layout annotations from VIGORv2, they capture more detailed object-level information, though with inevitable noise introduction. This trade-off between information richness and precision highlights the challenge of generating high-quality spatial semantic supervision from foundation vision models.



Figure 9: Comparative visualization of BEV transformation supervision information. This figure presents the supervision signals generated by our method, the BEV layout map from VIGORv2, and the corresponding satellite imagery.

A.2 Dynamic Sample Selection Strategy and Its Improvements.

During network training, we adopt the Dynamic Sample Selection (DSS) strategy to construct positive and negative samples. Specifically, DSS operates by running N training epochs over the dataset, after which the pairwise visual distances between all samples are computed and recorded using cosine similarity. This distance matrix is then used to generate the training sequence for the next epoch. For each query image, its top K nearest neighbors are selected to form the upcoming training batches. These K neighbors

are sorted by similarity: the top $\frac{k}{2}$ are chosen as the most similar samples, while the remaining $\frac{k}{2}$ are randomly sampled from the residual set. On the VIGOR dataset, the lack of geometric alignment significantly increases optimization difficulty, causing the feature space collapse observed in UMAP analysis (Fig.6), which renders the original DSS strategy ineffective. To address this issue, we introduce a modification to the DSS procedure: When the constructed training data under the current configuration covers less than 90% of the total dataset, we dynamically increase the value of K to expand the neighbor selection range. The successfully constructed training samples are then accumulated, and the process continues until a complete training sequence is formed. Following empirical insights from previous studies, we set $K=64$, $k=32$ and $N=4$ in our experiments to balance neighbor diversity and batch construction feasibility. For the BEV transformation task, maintaining the randomness of input batch sequences is critical to preventing overfitting. The DSS strategy addresses this by inserting randomly sampled instances into each batch, thereby achieving a balance between structural sample construction and stochasticity.

Table 6: RS refers to a random sampling strategy for constructing positive and negative pairs. DSS-Ori denotes the original Dynamic Similarity Sampling strategy from Sample4Geo. DSS-our represents our improved Dynamic Similarity Sampling strategy adapted for our setting.

Setting	SAME				
	R@1	R@5	R@10	R@1%	Hit Rate
RS	50.85	81.95	88.81	99.77	61.03
DSS-Ori	15.73	33.28	41.58	91.98	18.07
DSS-our	55.16	84.72	90.69	99.75	66.13

As shown in Table.6, the choice of sampling strategy significantly impacts model performance on the VIGOR dataset. Random sampling (RS) achieves 50.85% R@1, providing a baseline performance. Notably, random sampling remains effective because it does not rely on distance-based sample selection, thus avoiding the pitfalls of unreliable metrics caused by feature space collapse. In contrast, the original DSS strategy from Sample4Geo (DSS-Ori) performs poorly, achieving only 15.73% R@1—a 35.12 percentage point drop compared to random sampling. This dramatic degradation stems from the feature space collapse caused by lack of geometric alignment: collapsed features produce unreliable distance metrics, preventing DSS-Ori from identifying meaningful hard negatives and constructing stable training sequences. As shown in Fig.10, the DSS-Ori strategy exhibits severely fluctuating sequence lengths with drastically reduced training data volume. Our improved DSS strategy (DSS-our) mitigates this issue through dynamic K adjustment, achieving 55.16% R@1 and outperforming both random sampling by 4.31% and the original DSS by 39.43%, demonstrating the effectiveness of our adaptation in handling geometric misalignment challenges.

A.3 Analyzing the Effect of the w .

Our framework employs SSA as an auxiliary task to enhance the primary CVGL objective through spatial semantic alignment. The

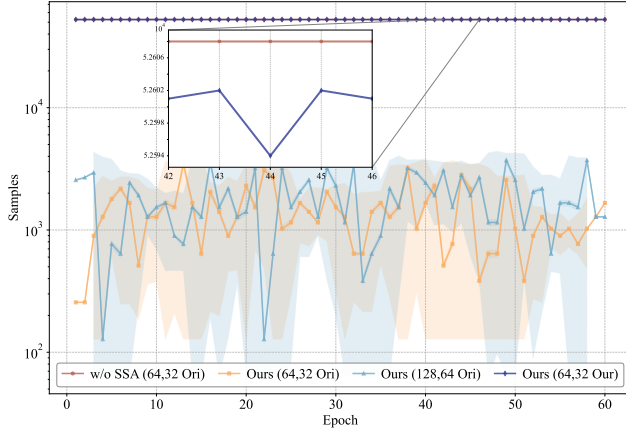


Figure 10: Visualization of training data sequence lengths under different settings. w/o SSA denotes SSA is removed, leaving only the contrastive learning. Ori denotes the original DSS strategy, while Our refers to our enhanced DSS strategy. The shaded region in the figure indicates the confidence interval.

Table 7: Effect of varying the weight w on final performance.

w	R@1	R@5	R@10	R@1%	Hit Rate
0.5	60.66	89.51	93.90	99.85	74.31
1	61.61	89.84	94.13	99.84	75.11
3	60.76	89.31	93.84	99.83	74.07

auxiliary SSA task aims to improve cross-view representation learning by providing additional spatial supervision signals through BEV transformation. While the primary goal remains cross-view geo-localization, the auxiliary task shares the same encoder architecture, enabling mutual information extraction and representation consistency across viewpoints. To investigate how the auxiliary SSA task influences the primary localization performance, we evaluate the model under varying weight parameter w values on the VIGOR dataset.

As shown in Table. 7, adjusting the auxiliary task weight w has minimal effect on the primary CVGL performance on VIGOR, with optimal results achieved at $w = 1$. The stability across different weight values indicates that the auxiliary SSA task is strongly coupled with the primary task through the shared encoder architecture. On VIGOR, the lack of geometric alignment forces the auxiliary SSA module to implicitly learn orientation correction while generating spatial semantic representations, leading to feature space collapse as observed in our UMAP analysis. This collapse creates optimization conflicts that propagate through the shared encoder regardless of the weight w , explaining why simply adjusting the auxiliary task weight cannot resolve the fundamental challenges posed by geometric misalignment. The phenomenon demonstrates that the auxiliary task’s influence on the primary objective stems not from gradient magnitude constraints, but from the high degree of mutual information sharing in the unified representation

learning framework, where failed auxiliary spatial learning due to geometric misalignment directly impacts the primary localization task through collapsed shared representations. Fig. 11 shows these prediction failures on VIGOR.

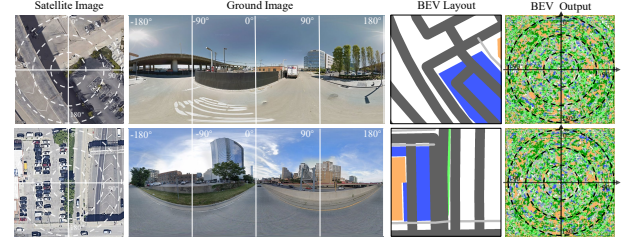


Figure 11: Visualization of failure cases on the VIGOR dataset. When using BEV layout annotations for supervision, demonstrating the difficulty of spatial semantic alignment without established geometric correspondence.

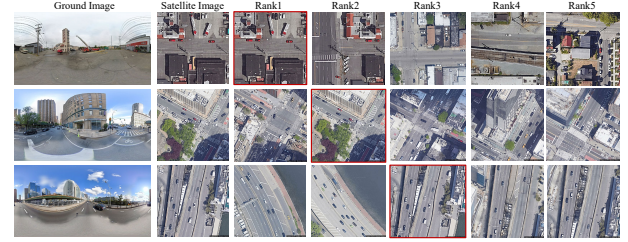


Figure 12: Visualization of image retrieval results. We present the retrieval results under the Same Area setting of the VIGOR dataset. Here, the rank indicates the reference images with the highest similarity scores. The correctly retrieved images are highlighted with red bounding boxes, providing an intuitive visualization of the model’s retrieval performance for this ground image.

A.4 Analysis of Failure Cases in Spatial Semantic Learning

CVUSA and CVACT apply geometric rectification with geographic north aligned to the top center, whereas VIGOR lacks this property. Ground-level panoramas in VIGOR have no consistent spatial orientation relative to satellite views and further include semi-positive samples with overlapping content but shifted capture locations. These dataset-specific geometric properties force the auxiliary SSA task to learn orientation correction and spatial alignment simultaneously, which substantially increases optimization difficulty. This leads to the feature space collapse observed in our UMAP analysis (Fig. 6). As shown in Fig. 11 and Fig. 12, the model retrieves visually plausible but geographically incorrect matches, and the predicted spatial semantic masks lose correct correspondence. This failure does not indicate a limitation of the SSA design itself, but rather arises from the lack of explicit geometric anchors and the inherent ambiguity in VIGOR. Without reliable geometric alignment, enriching semantic categories alone cannot resolve this issue.